

1 Question (10 points) (Algorithm Analysis)

Given the following function

```
int algorithm(int N){
    int sum = 0;
    if (N == 0)
        return 0;
    if (N % 2 == 0){
        for (i = 0; i < N; i++)
            for (int j = 0; j < N; j++)
                sum++;
    } else {
        for (int j = 0; j < N; j++)
            sum++;
    }
    return algorithm(N - 1) + sum;
}
```

What is the time complexity of the function algorithm? Show all your work. Derive the recursive equation and solve it.

2 Question (10 points) (Greedy)

There are n people who need to be assigned to execute n jobs, one person per job. (That is, each person is assigned to exactly one job and each job is assigned to exactly one person.) The cost that would occur if the i 'th person is assigned to the j 'th job is known quantity $C[i, j]$ for each pair $i, j = 1, 2, \dots, n$. The problem is to find an assignment with the minimum cost. You should return the assignment of jobs to persons as an array, so that `result[i]` shows the job index assigned to person i .

	Job 1	Job 2	Job 3	Job 4
Person 1	9	2	7	8
Person 2	6	4	3	7
Person 3	5	8	1	8
Person 4	7	6	9	4

```
int[] assignWithMinCost(int[] [] C)
```

3 Question (10 points) (Divide and Conquer)

You have an array A of n integers and another integer s . Find out whether the array contains four elements whose sum is s . Write a presorting algorithm of complexity $O(N^2 \log N)$. Hint: Think all possible pairwise sums.

```
boolean containsFourOfSumS(int[] A, int s)
```

4 Question (10 points) (Dynamic Programming)

Calculate the String number of the second kind using the following formula. Your algorithm should run in $O(N^2)$ time.

$$s(n, k) = s(n - 1, k - 1) - (n - 1)s(n - 1, k)$$

```
int stirlingOfSecondKind(int n, int k)
```

5 Question (10 points) (Brute Force)

Given an array A containing non-zero integers, determine the maximum range sum of A . In other words, find the maximum sum between any indices i and j in array A . Write a greedy algorithm to solve the problem in $O(N^3)$ time.

```
int maximumRangeSum(int[] A)
```